Project Application for the FISI Project in Learning Field 10

Class: 4o/4k
Project team: FiSi DV 14

1. Project Title

Automated Provisioning and Configuration of a Scalable Franchise Network Using Ansible

1.1 Short Description of the Task

Echt Hamburg GmbH plans to expand its existing business model to additional cities through a franchise system. Every franchise location requires a standardized network consisting of a router, a switch and a web server.

The purpose of this project is to design and implement the network of one pilot branch. Ansible will be used to configure the network devices remotely and automatically. The resulting solution should be reusable for future franchise locations by changing only site-specific variables such as the branch name, passwords, IP addresses and switch-port assignments.

1.2 Current-State Analysis

At the time of the project application, a router, a switch and a web server have already been delivered and physically installed at the pilot location. However, the devices have not yet been configured according to the company's technical requirements.

Without automation, the network devices of every new franchise location would have to be configured manually by qualified IT personnel. Repeating these tasks for every branch would require a significant amount of working time and could lead to inconsistent configurations and human error. Troubleshooting, documentation and the replacement of failed devices would also become more difficult as the number of franchise locations increases.

The internal network `192.168.108.0/24` and the public router address `200.108.1.1/28` have already been specified. The internal network must be divided into four VLANs for Customer Support, IT, the web server and device management.

2. Objectives / Development of the Target Concept

2.1 What Should Be Achieved at the End of the Project?

At the end of the project, a functional and documented pilot network for one franchise location will be available. The router and switch will be configured automatically using Ansible.

The internal network will be divided into four VLANs. The router will provide DHCP services for each VLAN and perform inter-VLAN routing. This will allow the departments to communicate while remaining logically separated.

The router and switch will be manageable through SSH using VLAN 99. NAT overload will be configured on the router so that internal private IPv4 addresses are not directly visible on the internet.

Unused switch ports will be disabled to reduce the risk of unauthorized network access. The router and switch will also display an MOTD banner containing the name of the franchise partner or project group.

A small website presenting the pilot branch and its team members will be deployed on the web server. In addition, a concept for monitoring the network devices and web server will be prepared.

The Ansible project will use reusable playbooks, roles and variables. This will enable future franchise networks to be deployed more quickly and consistently. Automation should reduce repetitive manual work, lower configuration costs, improve scalability and shorten recovery times after hardware replacement or configuration errors.

2.2 Which Requirements Must Be Fulfilled?

The following requirements must be fulfilled by the project:

- Creation of VLAN 10 for Customer Support
- Creation of VLAN 20 for IT
- Creation of VLAN 30 for the web server
- Creation of VLAN 99 for device management
- Division of `192.168.108.0/24` into suitable VLAN subnets
- Assignment of the first usable IPv4 address in each subnet to the router
- Dynamic IPv4 configuration in every VLAN using DHCP
- Inter-VLAN routing between the departments
- SSH management of the router and switch through VLAN 99
- Configuration of the domain `echt-hamburg.de`
- Configuration of the required unencrypted passwords
- Configuration of an MOTD banner containing the franchise or group name
- Deactivation of all unused switch ports
- NAT configuration for internet access
- Deployment of a small branch presentation website
- Creation of reusable and idempotent Ansible playbooks
- Documentation and testing of the complete solution
- Preparation of a basic monitoring concept
- Completion of an economic analysis comparing manual and automated deployment

2.3 Which Restrictions Must Be Considered?

Ansible requires an existing network connection and SSH access before it can configure a device. A minimal initial configuration may therefore have to be applied once through the console. All subsequent configurations will be performed through Ansible.

The public address `200.108.1.1/28` has already been specified. However, the external default gateway depends on the internet or laboratory environment and must be confirmed before implementation.

The task requires unencrypted passwords. This requirement will only be implemented in the isolated training environment. Plain-text passwords would not be recommended for a productive company network. For a production deployment, hashed passwords, Ansible Vault and centralized authentication should be used.
The project covers one pilot location only. High availability, redundant internet connections, wireless networking, IPv6 and the implementation of a central monitoring server are not included.

3. Project Planning

The project will be carried out as an in-house project. Short iteration cycles and regular feedback from the project team are planned. All Ansible files and documentation will be versioned using Git. The work steps and test results will be recorded in the project portfolio.

Analysis	**4h**
- Perform the current-state and requirements analysis	1h
- Examine the available hardware and software environment	1h
- Compare manual configuration, scripts and Ansible	1h
- Perform the economic feasibility analysis	1h
Design	**6h**
- Select the required technologies and Ansible modules	1h
- Create the network topology and VLAN design	1h
- Create the IPv4 addressing and DHCP plan	1h
- Develop the SSH, NAT and security concept	1h
- Prepare the project application	2h
Implementation	**24h**
- Prepare the physical or virtual laboratory environment	3h
- Install and prepare the Ansible control node	2h
- Create the inventory, variables, roles and playbooks	3h
- Configure basic router and switch settings, SSH and the MOTD banner	3h
- Configure VLANs, access ports and trunk connections	3h
- Configure router subinterfaces and inter-VLAN routing	2h
- Configure DHCP for all VLANs	2h
- Configure NAT/PAT and external connectivity	2h
- Disable and secure unused switch ports	1h
- Deploy the branch presentation website	1h
- Perform functional, connectivity and idempotency tests	2h
Documentation and Acceptance	**6h**
- Create the technical and administrator documentation	2h
- Develop and document the monitoring concept	1h
- Update the project portfolio	1h
- Prepare the project presentation	1h
- Conduct the final acceptance test and handover meeting	1h
Total:	**40h**